FS^{2}CCTrans: Frequency–Spatial–Spectral Joint Analysis With Criss-Cross Transformer for Hyperspectral Anomaly Detection

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***Abstract*— The distinguishability between background and anomaly is significant for accurate hyperspectral anomaly detection (HAD). The property that background and anomaly are characterized as signals with distinct differences in the frequency domain is of great value, however, existing HAD algorithms rarely consider this. In addition, deep learning (DL)-based methods that exploit reconstruction errors for HAD can inadvertently reconstruct anomalies alongside the background, leading to a high false alarm rate (FAR). To address these challenges, this study proposes a novel HAD model based on frequency–spatial–spectral domain analysis and criss-cross transformer (CCTrans), named FS^{2}CCTrans. Specifically, to suppress anomaly reconstruction, the frequency domain analysis paradigm is integrated into HAD, an advanced saliency map (SM) is constructed by analyzing the discriminative characteristics of anomaly and background in both amplitude and phase spectrum. The SM guides the model training in the direction of suppressing its capability to express and reconstruct anomalies, thereby obtaining a pure background estimate. To craft a superb background generator, the CCTrans network is designed that captures spatial–spectral features of the background in a very effective and efficient way. The CCTrans incorporates an ingenious criss-cross attention mechanism, which is focused on aggregating contextual information of all the pixels along its criss-cross path, thus significantly decreasing computational burden. Extensive comparisons with eight state-of-the-art methods on synthetic and real datasets demonstrate the superiority of FS^{2}CCTrans. Meanwhile, ablation studies reveal that the CCTrans requires about 68% less GPU memory and about 58% fewer FLOPs, highlighting its efficiency.**

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I. Introduction

HYPERSPECTRAL imaging (HSI) provides hundreds of contiguous spectral bands for each pixel. The rich spectral information enables precise identification of materials, which grants HSI unique advantages in detecting anomalous objects [1], [2], [3]. In complement to target detection techniques, hyperspectral anomaly detection (HAD) can identify anomalies by analyzing differences in spatial–spectral properties between background and anomaly without prior information [4], [5], making it essential for military reconnaissance [6], geological exploration [7], and agricultural monitoring [8]. Based on the strategy used to distinguish between background and anomalies, existing HAD methods can be classified into three groups: 1) statistical-based methods; 2) representation-based methods; and 3) deep learning (DL)-based methods.

Classical HAD methods are grounded in statistical theory. Among them, the RX detector holds a seminal position in the HAD literature [9], based on the assumption that the background follows a Gaussian multivariate distribution and detects anomalies using the Mahalanobis distance. However, this assumption may not always hold in practical scenarios. Building upon the RX detector, several advanced variants have been proposed, such as local RX [10] and weighted RX [11]. The two-stage GLRT (2S-GLRT) for multipixel anomaly detection (AD) was introduced in [12], incorporating background information from surrounding pixels. Recently, researchers have proposed detecting anomalies by analyzing HSI properties in the frequency domain. The MsRFQFT [13] investigates the characteristics of HSI in the frequency domain, revealing attributes like periodicity, frequency distribution, and energy concentration, which are not easily observable in the spatial–temporal domain.

Leveraging the robust capabilities of compressed sensing theory, representation-based methods have gradually emerged as the predominant approaches for HAD. These methods can be further categorized into low-rank representation (LRR) methods [14], [15], sparse representation (SR) methods

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[16], [17], and collaborative representation (CR) methods [18]. The GTVLRR [19] integrates graph and total variation (TV) regularizers with the LRR model to better assimilate spatial data from adjacent pixels. Some regularization terms, such as local spatial constraints and TV [20], and dual collaborative constraints regularized LR and SR [21], are incorporated into the LRR model. The CR model [22] introduces a dual-window strategy, where a pixel is considered part of the background if it can be linearly approximated by the pixels within its dual windows; otherwise, it is classified as an anomaly. To preserve the purity of the background dictionary, CRD with dual DR [23] and TV regularized LR decomposition [24] are developed. Despite their efficacy, representation-based methods typically require manual parameter tuning and are sensitive to parameter selection, which may lead to instability in detection.

DL has been instrumental in the advancement of HSI analysis, particularly in interpreting higher order information in complicated scenes [25]. Deep neural network (DNN), especially convolutional neural networks (CNNs) [26], autoencoders (AEs) [27], and generative adversarial networks (GANs) [28], exhibit exceptional abilities in mining hidden and nonlinear features, rendering them valuable tools for AD tasks. DL-based HAD models can be systematically classified into similarity measurement-based models and reconstruction-based models. Similarity measurement models utilize a CNN framework as the backbone and require the selection of predefined pixel pairs from labeled reference HSIs as training data [29]. The multilayer CNN is trained using the value difference between adjacent pixel pairs. In the evaluation phase, the trained CNN is employed to assess the differential values between adjacent pairs of pixels to identify anomalies. However, the prior information of objects is hard to obtain, and the transferability cannot be guaranteed. Conversely, reconstruction-based models circumvent the reliance on labeled datasets, thereby increasing their applicability. Given that background pixels predominate in HSIs, the model effectively learns background features, thus serving as an effective background reconstructor, and the reconstruction error forms the anomaly map.

Although reconstruction DNN can handle complex background modeling, some challenges have also arisen. First, most reconstruction-based models take the input HSI as the target for learning, leading the DNN model to treat anomalies and background equally. Consequently, as training progresses, anomalies may also be partially reconstructed, which reduces the purity of the reconstructed background and results in a high rate of missed detections (MDs). Second, AE-based methods are typically trained using spectral vectors, leading to inadequate exploration of spatial information. CNN is effective at extracting features within receptive fields but struggles with long-range pixel dependencies. Additionally, each pixel in the HSI contains tens to hundreds of spectral bands, presenting significant challenges in computational complexity and resource demands for processing such high-dimensional data.

Anomalies are fundamentally characterized as pixels exhibiting distinct spectral deviations from their local context, forming the discriminative basis of HAD. HSIs are essentially 2-D image arrays extended along the spectral axis. The transformer architecture [30], renowned for its capability to model long-range dependencies in sequential data, provides an ideal framework for extracting discriminative spectral–spatial features. Moreover, existing methodologies predominantly tackle HAD as the issue related to background modeling and AD in the spatial domain, often overlooking the frequency domain attributes. The frequency domain reveals the periodicity, frequency, and energy distribution of the data, and has been effectively applied in visual saliency detection [31], offering fresh insights into detection tasks. With these in mind, this study creatively introduces the frequency domain analysis paradigm into the HAD field. By thoroughly analyzing the features of HSIs in the frequency–spatial–spectral domain, a novel criss-cross transformer-based framework, F$\mathbf{S^2}$CCTrans, is proposed for HAD. Specifically, to suppress anomaly reconstruction, a saliency map (SM) is generated by exploiting the discriminative characteristics of anomaly and background in the frequency domain. This SM is then incorporated into the recurrent criss-cross transformer (RCCTrans) feature extractor to direct the training toward reconstructing only the background. To construct an efficient background reconstruction network, the criss-cross transformer (CCTrans) module is designed, which leverages the exceptional capabilities of the transformer in modeling long-range dependencies. Notably, the criss-cross attention (CCA) mechanism, originally developed for semantic segmentation [32], is adapted here for hyperspectral processing. This adaptation allows simultaneous modeling of spatial–spectral correlations while overcoming the computational complexity bottleneck inherent in conventional transformer designs. The synergistic integration of frequency-domain saliency guidance and efficient attention mechanisms establishes a new state-of-the-art in HAD performance. The main contributions of this study are summarized as follows.

1) We propose a novel CCTrans network to interpret the spatial–spectral characteristics in HSIs effectively. The CCTrans is based on the transformer architecture, which is inherently well-suited for HSI processing due to its ability to model long-range dependencies in spatial and spectral dimensions. Uniquely, the CCTrans considers the computational and memory challenges posed by the high dimensionality of HSI, and introduces a novel computational mechanism, criss-cross attention, which restricts the computation to pixels within the same row and column as the target pixel. This design substantially reduces computational complexity while equipping with advanced performance capabilities.
2) We develop a model-independent SM for anomaly suppression. Specifically, the original HSI is transformed into the frequency domain. A SM is constructed by analyzing the performance characteristics of anomalies and backgrounds within both the amplitude and phase spectrum, where the backgrounds are assigned high significance and the anomalies are opposite. This SM is then incorporated into the reconstructor's loss function, guiding the training process toward generating an

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accurate background estimation, thereby enhancing the separability of anomalies and backgrounds.

The remainder of this article is organized as follows. A review of related research is given in Section II. The methodology of the proposed FS²CCTrans is elaborated in Section III. Section IV details the experimental results and analysis, and finally, Section V provides a conclusion.

II. Related Work

A. Unsupervised DNNs for HAD

Depending on whether prior knowledge of anomaly targets is required, DNN-based HAD frameworks can be divided into supervised and unsupervised. Reconstruction-based methods directly take the unlabeled images as training data and extract the intrinsic features through unsupervised learning, making them more applicable to real-world scenarios. Common unsupervised DNN models include AEs [27] and GANs [28]. To leverage spatial context information and enhance network robustness, the robust graph autoencoder method (RGAE) [33] embeds graph regularization in the hidden layer of the autoencoder (AE) and redesigns the loss function. Zhang et al. [34] design the Auto-AD framework based on the full convolution of AE for efficient training and inference. Jung et al. [35] introduce the guided AE (GAED) to diminish the representation of the characteristics of the anomalies. Weak AD [36] employs a spectral-constrained GAN for HSI reconstruction, where anomalies exhibit greater reconstruction errors than backgrounds. Qu et al. [37] present a frequency-to-spectrum mapping GAN to enhance depth separable features of backgrounds and anomalies in the FrFD. More recently, transformers have attracted significant attention in the computer vision community. Xu et al. [38] introduced the vision transformer (ViT) model to the field of HAD for the first time and demonstrated its effectiveness in HAD tasks. Given that this study is based on the transformer, the basic principles are briefly described herein.

The transformer architecture follows an encoder–decoder design, where each encoder layer consists of the multihead self-attention (MSA) and the fully connected feed-forward network (FFN). The MSA is essentially a concatenation of several SA blocks. Specifically, for the hth head, the input $\mathbf{X}$ is mapped to the matrices $\mathbf{Q}$(query), $\mathbf{K}$(key), and $\mathbf{V}$(value) through three linear transformations (weight matrices). Attention scores are computed using the dot product of the query vectors and key vectors, followed by a softmax function to normalize the scores, resulting in weights:

$$\begin{aligned}\mathbf{Q}_h = \mathbf{X}\mathbf{W}_h^Q,\ \mathbf{K}_h = \mathbf{X}\mathbf{W}_h^K,\ \mathbf{V}_h = \mathbf{X}\mathbf{W}_h^V\\ \mathbf{Head}_h = \text{softmax}\left(\frac{\mathbf{Q}_h\mathbf{K}_h^T}{\sqrt{\mathbf{d}_k}}\right)\mathbf{V}_h\\ \mathbf{MultiHead}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Concat}(\mathbf{Head}_1,\ldots,\mathbf{Head}_h)\mathbf{W}^{\mathbf{O}}.\end{aligned} \tag{1}$$

Here, dividing by d_k aims to prevent the softmax function from entering regions with small gradients, and $\mathbf{W}^{\mathbf{O}}$ is the weight matrix.

B. Fourier Transform

The Fourier transform (FT) has been proven to be a powerful tool for HSI processing and is widely used in feature extraction [39] and target detection [40]. Given an image $\mathbf{X} \in \mathbb{R}^{M\times N}$, the 2-D FT is defined as

$$\mathbf{F}(u,v) = \sum_{n=0}^{N-1}\sum_{m=0}^{M-1}\mathbf{X}(m,n)e^{-2\pi i\left(\frac{mu}{M}+\frac{nv}{N}\right)} \tag{2}$$

where $\mathbf{F}$ is the frequency domain representation of $\mathbf{X}$, which contains both amplitude and phase information of the image. The amplitude reflects the intensity or energy of each frequency in the image. The mathematical representation is

$$\mathcal{A}(u,v) = \sqrt{\Re(\mathbf{F}(u,v))^2 + \Im(\mathbf{F}(u,v))^2} \tag{3}$$

where $\Re(\mathbf{F}(u,v))$ and $\Im(\mathbf{F}(u,v))$ represent the real and imaginary parts of $\mathbf{F}(u,v)$, respectively. The phase spectrum provides information about the relative positions of the frequency components, given as

$$\mathcal{P}(u,v) = \arctan\left[\frac{\Im(\mathbf{F}(u,v))}{\Re(\mathbf{F}(u,v))}\right]. \tag{4}$$

Then, the 2-D inverse FT (IFT) is defined as

$$\begin{aligned}\mathbf{X}(m,n) &= \frac{1}{MN}\sum_{v=0}^{N-1}\sum_{u=0}^{M-1}\mathbf{F}(u,v)e^{2\pi i\left(\frac{um}{M}+\frac{vn}{N}\right)}\\ &= \mathcal{F}^{-1}\left[\mathcal{A}(u,v)e^{i\mathcal{P}(u,v)}\right].\end{aligned} \tag{5}$$

III. Methodology

A. Overall Framework

The flowchart of FS²CCTrans is illustrated in Fig. 1. The framework consists of two branches: SM generation and model training. The SM is derived by analyzing the difference between the background and anomaly representations in both the amplitude and phase spectrum. It assigns a higher weight to the background while conferring the inverse weight to the anomalies, both of which are leveraged to facilitate network training. The designed model consists of three components: the preprocessing module, the RCCTrans module, and the reconstruction module. The preprocessing module is tasked with extracting local spatial–spectral features from the input and transforming them into a format that aligns with the input requirements of the RCCTrans module. This processed output is subsequently fed into the RCCTrans module to comprehensively extract and integrate global spatial–spectral features. Finally, the reconstruction module outputs the reconstructed background.

B. Preprocessing

Given an HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B correspond to the height, width, and spectral dimension, respectively. We use a sliding window to segment $\mathcal{X}$ into N overlapping HSI patches $\mathcal{X} = \{x_i\}_{i=1}^N$, and $x_i \in \mathbb{R}^{h\times w\times B}$. In this article, the sliding window size is set to 16×16 with a stride of 8. To enhance the interaction and integration of information across adjacent regions of the image while

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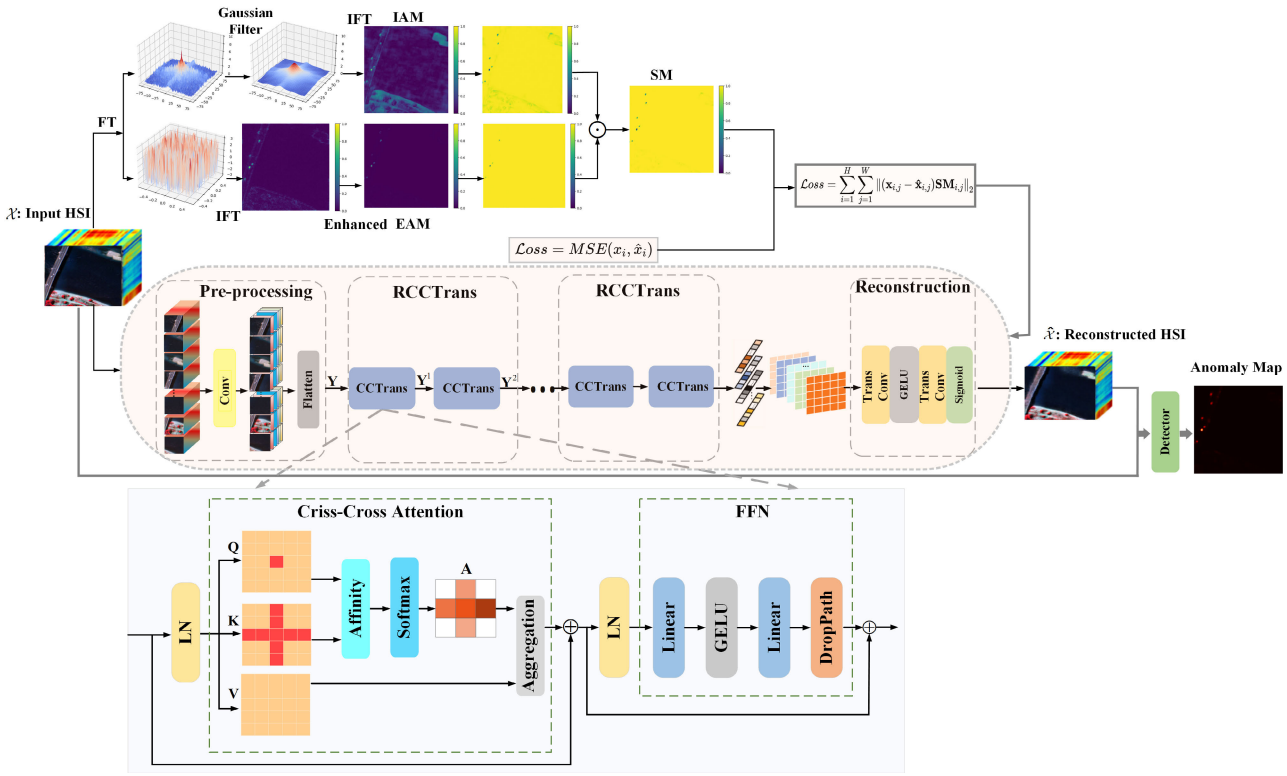


Fig. 1. Flowchart of FS^{2}CCTrans. The framework consists of two branches: SM generation (upper) and model training (lower). SM is obtained in the frequency domain. The amplitude and phase spectrum are processed by the respective upper and lower sub-branches to produce IAM and EAM components. These two components are then fused and postprocessed to derive the SM. The model training branch includes a preprocessing module, RCCTrans module, and reconstruction module. The loss function of the model is constructed by integrating the reconstruction error with the SM, and training under a specific strategy. Finally, the difference between the input and the reconstructed image is the anomaly map.

reducing the computational complexity of the model, these patches are mapped by a 3×3 convolution layer with an output channel number of C, which is set to 64 in this work. Each patch is then flattened into a vector by the flatten layer, and all vectors are used as input to the subsequent feature learning module.

C. Criss-Cross Transformer (CCTrans)

As illustrated in Fig. 1, the architecture of the CCTrans resembles the conventional transformer encoder, yet the attention mechanism is unique. We substitute the SA mechanism with criss-cross attention, which gathers contextual information through a criss-cross methodology. By applying recursive operations to the CCTrans block, the RCCTrans integrates spatial–spectral information locally and globally, resulting in rich global contextual information. In the constructed attention matrix, features with lower similarity are assigned lower weights, while highly similar features receive greater weights. These weighted features are crucial for effectively distinguishing background and anomalies.

For the given HSI $\mathcal{X}$, the preprocessing module is initially employed to derive a dimensionally reduced feature map $\mathbf{Y}$ that preserves local spatial information. $\mathbf{Y}$ is then fed into the first CCTrans to generate the feature map $\mathbf{Y^1}$. Based on the CCA mechanism, each pixel of $\mathbf{Y^1}$ can gather contextual information from the pixels along its criss-cross path. Notably, $\mathbf{Y^1}$ lacks the full global perspective information, which is insufficient for precise AD. To acquire a dense and extensive global feature, $\mathbf{Y^1}$ is processed again through the CCTrans, producing an updated feature map $\mathbf{Y^2}$. This allows each position within $\mathbf{Y^2}$ to aggregate the information from all pixels, achieving global spatial-spectral feature representation. This recursive architecture is termed RCCTrans, mathematically given as

$$\begin{aligned}\mathbf{Y^1} &= F_{\text{CCTrans}}(\mathbf{Y}) = \mathbf{Y} + f_{\text{CCA}}(LN(\mathbf{Y})) \\ &\quad + \text{FFN}\left[LN\left[\mathbf{Y} + f_{\text{CCA}}(LN(\mathbf{Y}))\right]\right] \qquad (6)\\ \mathbf{Y^2} &= F_{\text{CCTrans}}(\mathbf{Y^1}) = \mathbf{Y^1} + f_{\text{CCA}}\left(LN\left(\mathbf{Y^1}\right)\right) \\ &\quad + \text{FFN}\left[LN\left[\mathbf{Y^1} + f_{\text{CCA}}\left(LN\left(\mathbf{Y^1}\right)\right)\right]\right] \qquad (7)\end{aligned}$$

where $F_{\text{CCTrans}}(\cdot)$ is the function of the CCTrans block, $f_{\text{CCA}}(\cdot)$ denotes the function of the criss-cross attention, and $LN(\cdot)$ is the layer normalization process.

The RCCTrans architecture ensures that a pixel at any location within the input establishes dependencies with all other pixels. Fig. 2 illustrates the propagation of information from any given position $\mathbf{p^1}$ in the input feature $\mathbf{Y}$ to position $\mathbf{p^2}$ in the output feature $\mathbf{Y^2}$. Initially, the position $\mathbf{p^1}(p_x^1, p_y^1)$ conveys information through loop1 to the positions (p_x^2, p_y^1) and (p_x^1, p_y^2), which reside along the criss-cross trajectory of $\mathbf{p^2}$. Subsequently, these intermediary positions transmit the information to $\mathbf{p^2}(p_x^2, p_y^2)$ via loop2. This process ensures that even if $\mathbf{p^1}$ does not lie directly on the criss-cross path of $\mathbf{p^2}$, the information originating from $\mathbf{p^1}$ can still reach $\mathbf{p^2}$ by using two cascaded CCTrans blocks.

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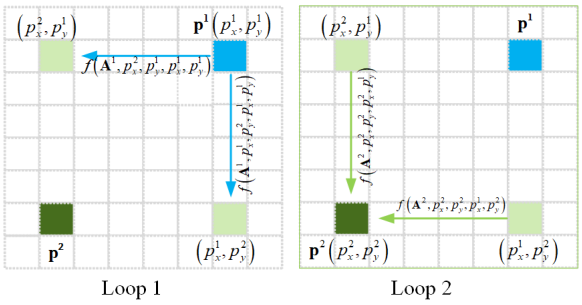


Fig. 2. Illustration of information propagation through two cascaded CCTrans Block.

1) Criss-Cross Attention: The criss-cross attention mechanism, as introduced in CCNet [32], is employed to efficiently capture global contextual information. This study adapts this mechanism to better align with the spatial–spectral characteristics of HSIs, thereby enhancing its effectiveness for AD. Specifically, for the feature map $\mathbf{Y}$, following the MSA mechanism, the feature is first linearly projected to h heads, each of which subsequently executes the CCA operation. For a single head, three feature maps $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are generated through three learnable matrices. The distinctive aspect of CCA lies in its affinity operation, which calculates attention scores along the criss-cross paths. This process unfolds in three distinct stages.

1) *Spectral vector extraction:* For the pixel at position p in the feature map $\mathbf{Q}$, a spectral vector $\mathbf{Q}_p$ is extracted.
2) *Formation of the set* $\mathbf{\Omega}_p$*:* From the feature map $\mathbf{K}$, spectral vectors corresponding to the same row and column as p are aggregated to form the set $\mathbf{\Omega}_p$ (ensuring that cross-position dependencies are captured without redundancy).
3) *Calculation of the attention score matrix* $\mathbf{S}$*:* The vector $\mathbf{Q}_p$ is multiplied with the set $\mathbf{\Omega}_p$ to derive the attention score matrix $\mathbf{S}$, which is subsequently normalized via a softmax function to generate the final attention map $\mathbf{A}$.

The affinity operation can be mathematically represented as follows:

$$\mathbf{s}_p = \mathbf{Q}_p \mathbf{\Omega}_p^{\mathrm{T}}. \tag{8}$$

Here, $\mathrm{s}_p \in \mathbf{S}$ indicates the correlation degree between $\mathbf{Q}_p$ and Ω_p.

Following the attention map, the aggregation function integrates contextual information from various regions of the image to form a global feature representation. For the position u of $\mathbf{V}$, the feature vectors in the same row and column as p are aggregated to construct $\mathbf{\Phi}_p$. This aggregation operation is mathematically expressed as

$$\mathbf{Y}_p^{\mathbf{1}} = \sum_{i \in |\mathbf{\Phi}_p|} \mathbf{A}_{i,p} \mathbf{\Phi}_{i,p} + \mathbf{Y}_p \tag{9}$$

where $\mathbf{Y}_p^{\mathbf{1}}$ denotes the feature vector at position p in $\mathbf{Y}^{\mathbf{1}}$. This process enhances the pixel-wise representation by adding contextual information to the local feature $\mathbf{Y}$, ensuring a comprehensive and contextually aware representation.

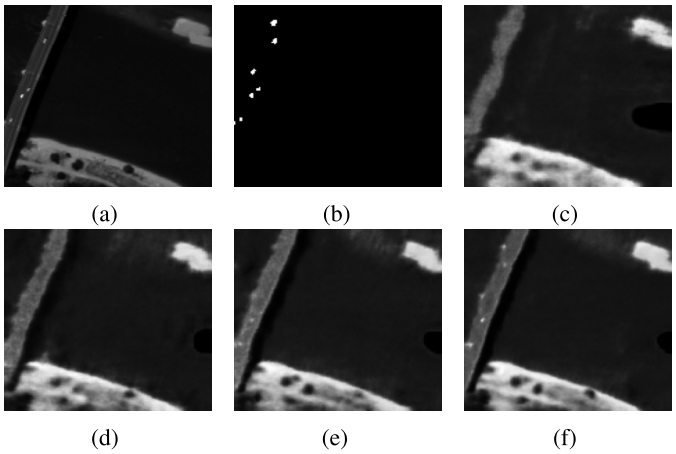


Fig. 3. Illustration of the overfitting problem (a) Input. (b) Ground-truth. (c) Iterations = 300. (d) Iterations = 400. (e) Iterations = 500. (f) Iterations = 600.

D. Reconstruction

After exhaustive analysis of the spatial–spectral features from both local and global of the input HSI, we designed a CNN-based image reconstruction module that maps the depth features back into the image space to obtain a reconstructed background. This unit contains two transposed convolution layers with kernel size 3×3 and stride 1. The first layer employs the GELU activation function to introduce nonlinearity, while the last layer utilizes the sigmoid function to constrain the pixel values of the output image within the range of 0–1.

E. Anomaly Suppression With Saliency Map

As discussed earlier, overfitting has become a significant risk in reconstruction-based HAD methods. This is mainly due to the equal treatment of input pixels by the model, which, as the training process advances, enhances the network's capability to fit the data and tends to reconstruct the anomalies. Fig. 3 illustrates this. During the early stage of network training (iterations = 300), the model focuses mainly on reconstructing the background. As training continues (iterations = 500), background reconstruction improves while anomaly targets also begin to be reconstructed. By iterations = 600, the anomalies are reconstructed as part of the background. Therefore, maintaining the integrity of the background while suppressing anomalies during training is a major challenge. The frequency of an image serves as an indicator to evaluate the degree of change within the local regions, which is highly beneficial for the identification of anomalies, hence, this study introduces the frequency domain analysis paradigm to the HAD task.

1) Amplitude Spectrum Analysis for IAM: In [31], a novel approach for visual saliency analysis was proposed by performing scale space analysis on the amplitude spectrum of natural images. This study revealed an intrinsic rule: for natural images, backgrounds with smooth, periodic repetitive patterns concentrate their amplitude spectrum energy at low frequencies, which results in the background behaving as spikes. Consequently, smoothing the amplitude spectrum

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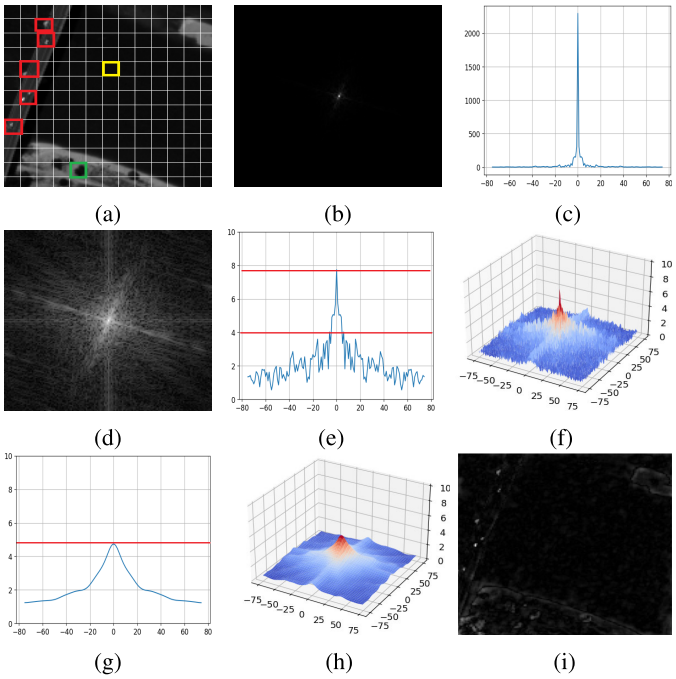


Fig. 4. Examples of periodic backgrounds corresponding to spikes in the amplitude spectrum. (a) Input image. (b) Amplitude spectrum. (c) 1-D form of (b). (d) Log amplitude spectrum. (e) 1-D form of (d). (f) 3-D form of (d). (g) 1-D form of filtered amplitude. (h) 3-D form of filtered amplitude. (i) IAM.

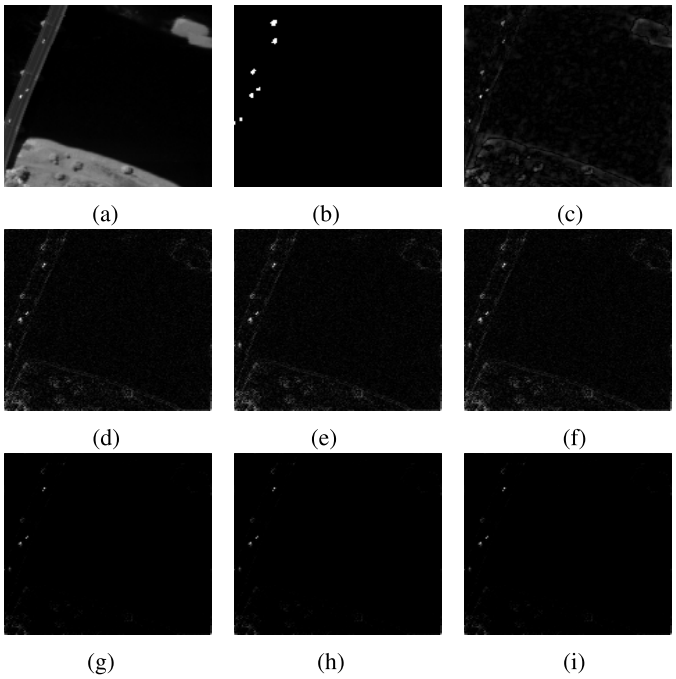


Fig. 5. Examples of reconstruction using the POR. (a) Input. (b) Ground truth. (c) Initial anomaly. (d) $\mathcal{A}_{\text{unit}} = 1$. (e) $\mathcal{A}_{\text{unit}} = 100$. (f) $\mathcal{A}_{\text{unit}} = 1000$. (g) Square operation of (d). (h) Square operation of (e). (i) Square operation of (f).

with a Gaussian kernel and reconstructing the signal with the original phase spectrum is equivalent to the saliency detector. After careful analysis, we find that the amplitude spectrum of the HSI also follows this pattern. In Fig. 4, the widely used HSI (Pavia) is adopted to illustrate this. Given the extensive dynamic range of the energy distribution in the amplitude [Fig. 4(c)], it is difficult to identify and analyze high-frequency information (which corresponds to image edges and details). Therefore, we apply a logarithmic transformation to the amplitude: $\mathcal{L}(u, v) = \log(1 + \mathcal{A}(u, v))$. Fig. 4(e) and (f) shows the 1-D and 3-D representations of the log amplitude, respectively. It is observed that energy across different frequency components appears as several spikes, with the most prominent spike near zero frequency, corresponding to the background with the highest recurrence rate [marked by yellow boxes in Fig. 4(a)]. Other spikes correspond to anomalous targets (marked by red boxes) and segments of the background (marked by green boxes). To highlight the high-frequency components of the anomalies, a Gaussian low-pass kernel $\mathcal{G}_\sigma$ with a standard deviation of σ is utilized to suppress high-energy spikes in the amplitude[1]

$$\mathcal{S}_{\mathcal{A}}(u, v) = \mathcal{L} * \mathcal{G}_\sigma \tag{10}$$

where $\mathcal{S}_{\mathcal{A}}(u, v)$ is the filtered amplitude, and $*$ represents the convolution operation. Combined with the original phase spectrum $\mathcal{P}(u, v)$, the initial anomaly map (IAM) can is obtained through the IFT

$$\mathbf{IAM} = \left| F^{-1}\left[S_A(u, v)e^{i\,P(u,v)}\right]\right|. \tag{11}$$

[1]One might mistakenly think that this Gaussian low-pass filter is equivalent to removing high-frequency information in the spatial domain. Yet, this is not the case, as we perform on the amplitude, removing its drastically varying part without changing the original phase.

Fig. 4(g) and (h) shows the 1-D and 3-D forms of the amplitude spectrum after low-pass filtering ($\sigma = 3.6$), where the energy peaks of the low-frequency components are significantly suppressed. Fig. 4(i) presents the IAM with the anomalies highlighted.

2) Phase Spectrum Analysis for Enhanced Anomaly Map: Observation of Fig. 4(i) reveals that some background regions are also highlighted (lower left and upper right of the image). Therefore, further refinement is necessary to suppress the nonanomaly, nonperiodic changing regions. The amplitude spectrum reflects the number of each periodic component, while the phase spectrum specifies their positions. Regions with low periodicity or significant variations manifest as emphasized targets in the phase-only reconstruction (POR), revealing potential candidate targets [13]. As discussed earlier, anomalies are characterized by their sparsity, locality, and irregularity. Sparsity reflects the low periodicity of anomalies, while locality and irregularity correspond to anomalies manifesting as small, highly variable regions. Thus, using POR to identify candidate targets is well-suited for HAD. Specifically, when applying the Gaussian low-pass filter to smooth the energy peaks in the amplitude spectrum, the value of σ determines the degree of smoothing. If σ is large enough, the filtered amplitude will become homogeneous, where the amplitude values of all frequency components are nearly equal, leaving the image reconstruction dependent solely on the phase information. To obtain an enhanced anomaly map, we adopt the POR method, and the mathematical representation is as follows:

$$\mathbf{EAM} = \left| \mathcal{F}^{-1}\left[\mathcal{A}_{\text{unit}} e^{i\mathcal{P}(u,v)}\right]\right|^2 \tag{12}$$

where $\mathcal{A}_{\text{unit}}$ denotes the uniform amplitude spectrum. Fig. 5 shows the phase reconstruction results for varying values

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of $\mathcal{A}_{\text{unit}}$. It is observed that an increase in $\mathcal{A}_{\text{unit}}$ from 1 to 1000 results in minimal changes in output image brightness and contrast, with negligible impact on structural features. This is because the uniform amplitude spectrum lacks spikes, thereby emphasizing high-frequency components. The phase spectrum, on the other hand, contains the spatial structure of the image, making anomalies more prominent. For computational simplicity, the value of $\mathcal{A}_{\text{unit}}$ is set to 1 in this study. To further enhance anomaly visibility, a squaring operation is applied to the phase reconstruction, effectively increasing the contrast between high-brightness areas (corresponding to anomalies) and the background. As shown in the third row of Fig. 5, the anomalies become more discernible after the squaring operation, aiding in subsequent analysis and identification.

3) Reconstruction Fusion for Saliency Map: The anomalies may present as variations in intensity, texture, or spatial structure, thus, we integrate IAM and EAM to utilize complementary information and diminish the false positives that rely solely on one type of information. Given an HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$ and $\{\mathbf{X}_k\}_{k=1}^{B} \in \mathbb{R}^{H\times W}$ as its set of bands. For the 2-D image $\mathbf{X}_k$, it is first transformed into the frequency domain. The IAM $\mathbf{IAM}_k \in \mathbb{R}^{H\times W}(k = 1, 2, \ldots, B)$ and enhanced anomaly map $\mathbf{EAM}_k \in \mathbb{R}^{H\times W}(k = 1, 2, \ldots, B)$ are calculated according to (11) and (12), respectively. Then, $\mathbf{IAM}_k$ and $\mathbf{EAM}_k$ for all bands are aggregated and collapsed to produce the final $\mathbf{IAM} = \sum_{k=1}^{B} |\mathcal{F}^{-1}[\mathcal{S}_{Ak}(u, v)e^{i\mathcal{P}_k(u,v)}]|$ and $\mathbf{EAM} = \sum_{k=1}^{B} |\mathcal{F}^{-1}[\mathcal{A}_{\text{unit}}e^{i\mathcal{P}_k(u,v)}]|^2$. For **IAM** and **EAM**, after normalization, the anomaly values are larger and closer to 1, while the background pixels are in the opposite direction. The SM can be expressed as

$$\mathbf{SM} = \text{Sigmoid}(\mathbf{J}_{H\times W} - \mathbf{IAM}) \odot \text{Sigmoid}(\mathbf{J}_{H\times W} - \mathbf{EAM}). \tag{13}$$

Here, $\odot$ denotes the element-wise product, and $J_{H\times W}$ represents a matrix with all elements equal to 1. To further enhance the distinction between anomalies and background, we apply the sigmoid function for nonlinear mapping to the difference result. Prior to applying the sigmoid function, the mean values of the two difference maps are adjusted to zero, followed by normalization to the range $[-v_s, v_s]$. This ensures that most elements fall within the sensitive range of the sigmoid function. As illustrated in Fig. 6(c) and (e), the results after sigmoid processing demonstrate that the anomalies are further accentuated. Finally, the SM is generated, where background pixels are assigned higher weights, while anomalies are assigned weights close to zero, as depicted in Fig. 6(f).

4) Anomaly Suppression: To suppress the reconstruction of anomalies, the SM is embedded into the proposed model. Specifically, we combine the SM with the original loss function and update the network by backpropagation. Since the weights of anomalies are small, their contribution to the total loss is significantly reduced, effectively constraining their reconstruction and facilitating the learning of the pure background. The final loss function of the model is given by

$$\mathcal{L}\text{oss} = \sum_{i=1}^{H}\sum_{j=1}^{W} \left\| (\mathbf{x}_{i,j} - \hat{\mathbf{x}}_{i,j})\mathbf{SM}_{i,j} \right\|_2 \tag{14}$$

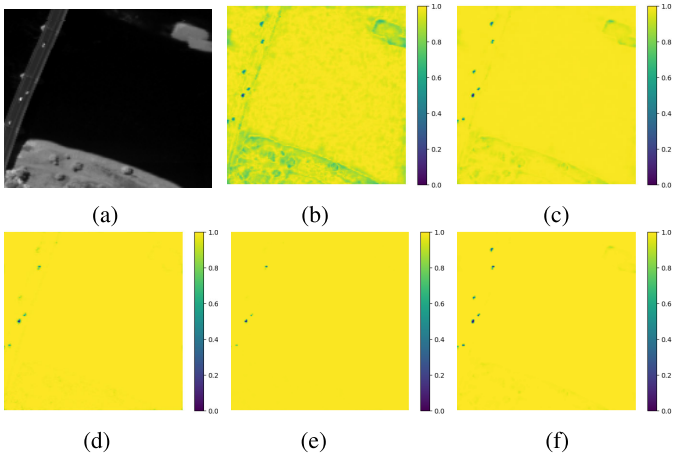


Fig. 6. Examples of reconstruction fusion for SM. (a) Input. (b) Difference of IAM. (c) Sigmoid of (b). (d) Difference of EAM. (e) Sigmoid of (d). (f) SM.

where $\mathbf{x}_{i,j} \in \mathbb{R}^{B\times 1}$ and $\hat{\mathbf{x}}_{i,j} \in \mathbb{R}^{B\times 1}$ represent the pixel vectors at location (i, j) of the input image $\mathbf{X}$ and the reconstructed background image $\hat{\mathbf{X}}$, respectively. $\mathbf{SM}_{i,j}$ denotes the weight at location (i, j).

The SM aims to mitigate the impact of anomalies on cumulative loss, thereby averting the reconstruction of anomaly targets. In this study, the SM was mainly involved in the first 400 iterations of training, with model adjustments occurring every two iterations. The ADAM algorithm [41] is adopted to update the parameters of the network. Training will be stopped when the average loss change over the last 50 iterations falls below $\epsilon = 1.5 \times 10^{-5}$

$$\frac{1}{50}\sum_{n=k}^{k+50}\left(\mathcal{L}\text{oss}^{n+1} - \mathcal{L}\text{oss}^{n}\right) < \epsilon \tag{15}$$

where k denotes the iteration count. After training is finished, the output image of the model is the reconstructed background. The anomaly map is then obtained by computing the difference between the input and the reconstructed background. The final detection map $\mathbf{D} \in \mathbb{R}^{H\times W}$ is computed as

$$\mathbf{D} = \mathcal{F}\left(\|\mathbf{x}_{i,j} - \hat{\mathbf{x}}_{i,j}\|^2\right) \tag{16}$$

where $\mathcal{F}(\cdot)$ is the fold operation along the band dimension.

IV. Experiments and Analysis

This section presents a comprehensive evaluation of the proposed FS²CCTrans. First, we describe the experimental setup, including the datasets, baseline methods, and evaluation metrics. Next, we provide a detailed analysis of the experimental results. Finally, we conduct ablation studies to validate the effectiveness of key components in the proposed framework.

A. Datasets Description

1) Synthetic Dataset: The first dataset is a synthetic dataset generated using HSIs captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). Specifically, a 120 × 120 subset is selected as the background image, and 25 anomaly targets are implanted using the target implantation

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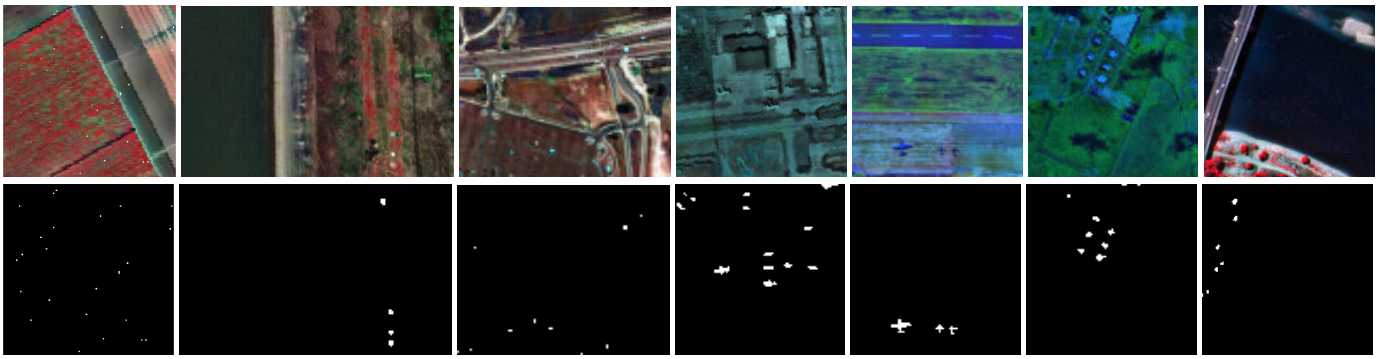


Fig. 7. Pseudo-color images and ground-truth images of seven datasets. (a) Synthetic dataset. (b) WHU-Hi-River. (c) HYDICE. (d) Los Angeles-1. (e) Gulfport. (f) Texas Coast. (g) Pavia.

TABLE I
DETAILS OF THE REAL DATASETS

Datasets	image size ($H \times W \times B$)	Resolution	Ratio
WHU-H-River	$105 \times 168 \times 135$	0.06 m/pixel	0.20%
HYDICE	$80 \times 100 \times 162$	1 m/pixel	0.20%
Los Angeles-1	$100 \times 100 \times 205$	7.1 m/pixel	1.44%
Gulfport	$100 \times 100 \times 191$	3.4 m/pixel	0.60%
Texas Coast	$100 \times 100 \times 204$	17.2 m/pixel	0.67%
Pavia	$150 \times 150 \times 102$	1.3 m/pixel	0.30%

method proposed by Stefanou and Kerekes [42]. This method is based on the linear mixed model

$$z_{\text{spe}} = f_{\text{abu}} t_{\text{spe}} + (1 - f_{\text{abu}}) b_{\text{spe}} \tag{17}$$

where z_{spe} represents the synthesized spectral signature, while t_{spe} and b_{spe} correspond to the spectrum of the anomaly and background, respectively. The abundance fraction f_{abu} varies from 0.04 to 1 in increments of 0.04. Fig. 7(a) presents the pseudo-color image and ground truth of the synthetic dataset.

2) Real Dataset: The remaining six datasets are all from real scenarios, specifically WHU-Hi-River, HYDICE, Los Angeles-1, Gulfport, Texas Coast, and Pavia City. The WHU-Hi-River dataset was collected by the RSIDEA team at Wuhan University using a Nano Hyperspec sensor mounted on an uncrewed aerial vehicle (UAV), where two plastic plates and two gray plates serve as anomalies.[2] The HYDICE dataset was collected by the Hyperspectral Digital Imagery Collection Experiment sensor over a suburban residential area in Michigan, USA, with ten vehicles considered as anomalies. The latter four datasets are publicly available online.[3] Table I provides a detailed description of these datasets. Additionally, Fig. 7 shows the pseudo-color images and the ground truth.

B. Experiment Setup

1) Comparison Methods: Eight state-of-the-art HAD methods are selected for comparative analysis, which includes the statistical methods RX [9], 2S-GLRT [12], MsRFQFT [13], the representation-based methods CRD [22], GTVLRR [19], and the DL-based methods Auto-AD [34], RGAE [33], S2DWMTrans [35]. Among these, Auto-AD designs a framework of multiple convolutional layers, while S2DWMTrans pioneers the integration of ViT into HAD tasks. These DL-based approaches represent cutting-edge advancements in DL for AD. To ensure fairness, optimal parameters are chosen for the compared methods. For CRD and 2S-GLRT, the window sizes w_{out} (5 to 25) and w_{in} (3 to 15) are adjusted for optimal performance. For MsRFQFT, the selected band number N_d is fixed at 3, with the Gaussian kernel scale σ tuned within [0.2,10] at 0.2 intervals. For GTVLRR, dictionary construction is implemented via K-means clustering, with initial regularization parameters $(\lambda, \beta, \gamma) = (0.5, 0.2, 0.05)$. Optimal parameters are determined through a sequential optimization strategy: fixing two parameters while varying the third. The Auto-AD operates autonomously without requiring additional parameter tuning. For RGAE, the configuration includes: random weight and bias initialization; a tradeoff parameter of 0.01; a hidden layer dimension of 100; 150 superpixels; a maximum iteration limit of 300; a learning rate of 0.01; and a final iteration limit of 1200. For S2DWMTrans, optimal window sizes are set to $w_{\text{in}} \in [3, 7]$ following the original implementation.

For the proposed FS^{2}CCTrans, the initial learning rate is set to 1×10^{-4}, and a cosine annealing learning rate scheduler with a 100-epoch linear warm-up phase is employed. The batch size is configured to match the total number of pixels in the input HSI. The model is trained for 800 epochs to ensure complete convergence. In terms of architecture, the preprocessing module is designed with an output dimension of 128 features, while the spectral vector is compressed to 64 dimensions. The RCCTrans module incorporates two fully connected layers in its feedforward networks, with the first and second layers containing 256 and 128 units, respectively. Notably, the number of attention heads in the CCA block is set to 2, a configuration that effectively balances the model's capacity to handle data complexity with computational efficiency. All experiments are conducted on a workstation equipped with an Intel Core i7-12700KF CPU (3.6 GHz, 32 GB RAM) and an NVIDIA GeForce RTX 3060 GPU, utilizing Python 3.8, PyTorch 1.13, and MATLAB R2021b as the software environment.

[2]http://rsidea.whu.edu.cn/resource_WHUHiriver_sharing.htm
[3]http://xudongkang.weebly.com

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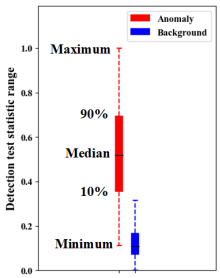
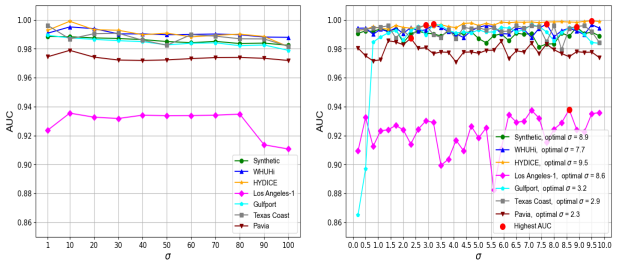


Fig. 8. Illustration of the box-whisker plot.

2) Evaluation Metrics: The performance of these detectors is evaluated from both qualitative and quantitative perspectives. Qualitative metrics include the color-coded anomaly map, the box-and-whisker plot, and the receiver operating characteristic (ROC) curve [43]. The area under the ROC curve (AUC) serves as the primary quantitative metric. Specifically, the color anomaly maps provide visual representations of detection effectiveness, highlighting both the response intensity of anomalies and the suppression degree of background pixels. The box-and-whisker plots illustrate the statistical distribution of detection scores, demonstrating each detector's capability to discriminate between background and anomalous pixels. As shown in Fig. 8, the red box indicates the distribution of anomalies, while the blue box represents the background pixels. Notably, the greater the distance between the red and blue boxes, the better the separability of the detector. Furthermore, a reduced height of the blue box indicates enhanced background suppression, which corresponds to a lower false alarm rate (FAR). The ROC curve depicts the relationship between the probability of detection (PD) and the FAR at different thresholds (τ). A curve closer to the upper left corner of the plot indicates better performance of the detector.

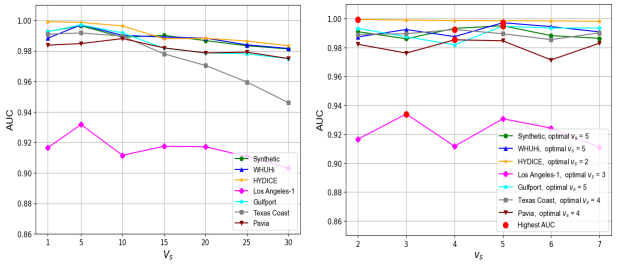
C. Parameter Analysis

1) Standard Deviation σ: The parameter σ controls the filtering scale of the amplitude spectrum. A small σ fails to fully eliminate spikes, leading to background artifacts in the IAM. As σ increases, the amplitude spectrum goes smoother and suppresses the background better. However, excessively large values may inadvertently suppress the anomalies. To identify the optimal filtering scale, we conducted systematic experiments across seven datasets using a coarse-to-fine parameter selection strategy. As illustrated in Fig. 9(a), we first assessed the overall trend of AUC values over an extensive parameter range ($\sigma \in [1, 100]$) with v_s fixed at 4. The results indicate an increasing AUC as σ rises from 1 to 10, beyond which the AUC stabilizes or slightly declines. For instance, peak AUC values of 0.997 and 0.995 are achieved at $\sigma = 7.7$ (WHUHi) and $\sigma = 8.9$ (Synthetic), respectively. These findings confirm that $\sigma \in [1, 10]$ constitutes the optimal operational range. Subsequently, we conducted a refined analysis within this range to evaluate the robustness and sensitivity of the proposed method. Fig. 9(b) reveals that for six datasets (excluding Los Angeles-1), AUC values remain stable with a maximum fluctuation of 0.015, indicating robustness to parameter



(a) (b)

Fig. 9. Effect of the standard deviation σ on the performance of FS²CCTrans. (a) AUC trend across a broad range of σ. (b) AUC fluctuations within the optimal range of σ.



(a) (b)

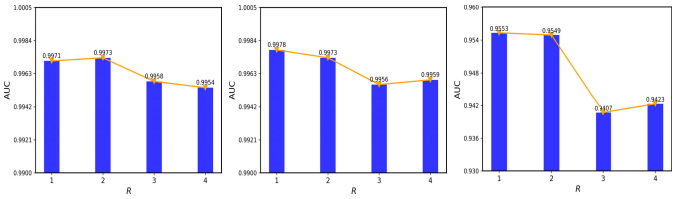
Fig. 10. Effect of the normalized range V_s on the performance of FS²CCTrans. (a) AUC trend across a broad range of V_s. (b) AUC fluctuations within the optimal range of V_s.

changes. The optimal σ values for the FS²CCTrans across the seven datasets are set to $\{8.9, 7.7, 9.5, 8.6, 3.2, 2.9, 2.3\}$.

2) Normalized Range v_s: The sigmoid function is utilized to perform a nonlinear mapping of the differences of the IAM and EAM, respectively. This operation is designed to further enhance the separability between the background and anomalies. To determine the optimal v_s value, a coarse-to-fine manner similar to the analysis of σ was employed. First, a comprehensive evaluation of v_s was undertaken within the range $[1, 30]$, while keeping σ at its optimal configuration. As illustrated in Fig. 10(a), the AUC initially improves with increasing v_s, reaching peak performance within $v_s \in [1, 8]$, beyond which performance stabilizes or marginally degrades. This suggests that the optimal v_s resides within this specified range. Following this, a more meticulous examination was carried out within this range, as presented in Fig. 10(b), which demonstrates minimal AUC fluctuations across seven datasets as v_s changes, with HYDICE notably stable and maintaining high AUC values. The performances affirm that the proposed FS²CCTrans is robust against variations in v_s, ensuring dependable AD. Based on the optimal AUC, the v_s of the FS²CCTrans on the seven datasets is set to $\{5, 5, 2, 3, 5, 4, 4\}$.

3) Number of RCCTrans R: The depth parameter R (number of recurrent blocks) critically influences the network's representational capacity. An excessively shallow network cannot adequately capture abstract semantic features, whereas excessive depth may result in overfitting. To systematically analyze this trade-off, we evaluated $R \in \{1, 2, 3, 4\}$ across three representative datasets, monitoring both detection performance (AUC) and computational complexity (FLOPs).

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(a) (b) (c)

Fig. 11. Effect of the network depth R on the performance of FS^{2}CCTrans. (a) Gulfport. (b) Texas Coast. (c) Los Angeles-1.

TABLE II
FLOPs (BILLION) INCREMENT ON THREE DATASETS FOR DIFFERENT NUMBER OF LOOPS R

Loops	$R=1$	$R=2$	$R=3$	$R=4$
Gulfport	18.33	36.65	54.97	73.32
Texas Coast	19.10	38.1	57.4	76.4
Los Angeles-1	18.09	36.16	54.24	72.32

As summarized in Fig. 11 and Table II, it was noted that only the Gulfport dataset exhibited a slight improvement in AUC R was increased from 1 to 2, a further increase in depth negatively impacted performance across all datasets. Meanwhile, the FLOPs escalated linearly with the increase in loop numbers. To balance the performance and resource usage, we set $R = 1$ as the default setting for subsequent experiments.

D. Experimental Result

1) Qualitative Evaluation: Fig. 12 presents the color anomaly maps on the seven datasets, where column 1 is the ground truth and subsequent columns illustrate detection results from various methods. The FS^{2}CCTrans achieves an optimal trade off between precise anomaly identification and rigorous background suppression.

For the synthetic dataset (column 2), the RX performs poorly. Although CRD and GTVLRR can identify certain anomalies, they also misclassify some backgrounds as anomalies, with GTVLRR in particular exhibiting a high FAR. 2S-GLRT can highlight the anomalies; however, it can be seen that the boundaries of each target contain some background structures, resulting in detected targets appearing larger than their actual size. The MsRFQFT, Auto-AD, RGAE, and S2DWTrans detectors also exhibit background suppression, but the proposed FS^{2}CCTrans has a more pronounced anomalous response to embedded pixels, suggesting that FS^{2}CCTrans performs better at separating backgrounds and anomalies.

For real-world scenes, taking HYDICE, Los Angeles-1, and Gulfport as examples, which have complex background distributions, FS^{2}CCTrans demonstrates superior performance in both anomaly enhancement and background suppression. Although RX, CRD, GTVLRR, and 2S-GLRT successfully identify some anomalies, part of the background structures are also misclassified as anomalous, making it difficult to distinguish them visually. The MsRFQRT and Auto-AD show satisfactory results for background suppression in the HYDICE and Los Angeles-1 image, but some anomalies are lost. RGAE performs better than S2DWTrans in background suppression, but its effect on anomaly localization is not ideal. Furthermore, the results of the proposed FS^{2}CCTrans converge with the ground truth on the WHU-Hi-River, Texas Coast, and Pavia datasets. Judging from the HAD color images, our proposed FS^{2}CCTrans shows relatively excellent visual effects.

Fig. 13 presents the AD results for all methods in the form of box-and-whisker plots. To facilitate comparative analysis, the detection values are normalized to [0,1]. The box-and-whisker plots intuitively reflect the performance of different detectors in terms of background suppression and anomaly-background separability. The greater the distance between the background and anomaly boxes, the better the detector's ability to distinguish between the two. It can be observed that FS^{2}CCTrans achieves clear separation between the background and anomaly boxes across all datasets, with the distance between the two boxes ranking among the best. While GTVLRR and CRD also show a notable gap, the upper whiskers of the background box for the two methods are too high compared to FS^{2}CCTrans, indicating they are less effective in suppressing the background. Furthermore, the blue boxes of FS^{2}CCTrans are concentrated at a very low level, especially the middle 80% of the background pixels, indicating its excellent background suppression. Taking Los Angeles-1 as an example, as shown in Fig. 13(d), except for FS^{2}CCTrans, the blue background boxes of all other detectors are located higher, indicating inadequate background suppression, while FS^{2}CCTrans is still working. In summary, FS^{2}CCTrans shows excellent performance in background-anomaly separability and background suppression. Fig. 14 presents the corresponding ROC curves. In most cases, the ROC curve of FS^{2}CCTrans surpasses those of the other methods, staying closest to the upper left corner of the axis. On the HYDICE dataset, although FS^{2}CCTrans exhibits a slightly lower PD value than CRD, 2S-GLRT, GTVLRR, and Auto-AD when the PF value is below 0.11, the overall comparison shows that FS^{2}CCTrans maintains a higher PD. When PD reaches 100%, the PF values of FS^{2}CCTrans, RGAE, Auto-AD, S2DWTrans, GTVLRR, CRD, MsRFQFT, 2S-GLRT, and RX are approximately 0.8%, 9%, 5%, 7%, 1%, 2.5%, 6.5%, 3%, and 0.6%, respectively. For the other six datasets, FS^{2}CCTrans achieves the lowest PF values at 100% PD, indicating that the SM effectively suppresses the reconstruction of anomalies.

2) Quantitative Evaluation: Table III lists the corresponding AUC scores, with the highest values highlighted in bold and the suboptimal ones underlined. The FS^{2}CCTrans outperforms all other methods across all seven datasets, with an average value of 0.9919. Notably, FS^{2}CCTrans delivers a 3% absolute AUC improvement (0.9553 versus 0.9236) over the second-best method (2S-GLRT) on the challenging Los Angeles-1 dataset, which features complex background distributions and a high anomaly concentration. This significant performance gap underscores the method's enhanced capability in handling intricate detection scenarios through joint frequency–spatial–spectral feature analysis. While the performance gains over Auto-AD (the suboptimal baseline) appear less pronounced on five other datasets, this observation aligns with our design objective: FS^{2}CCTrans prioritizes

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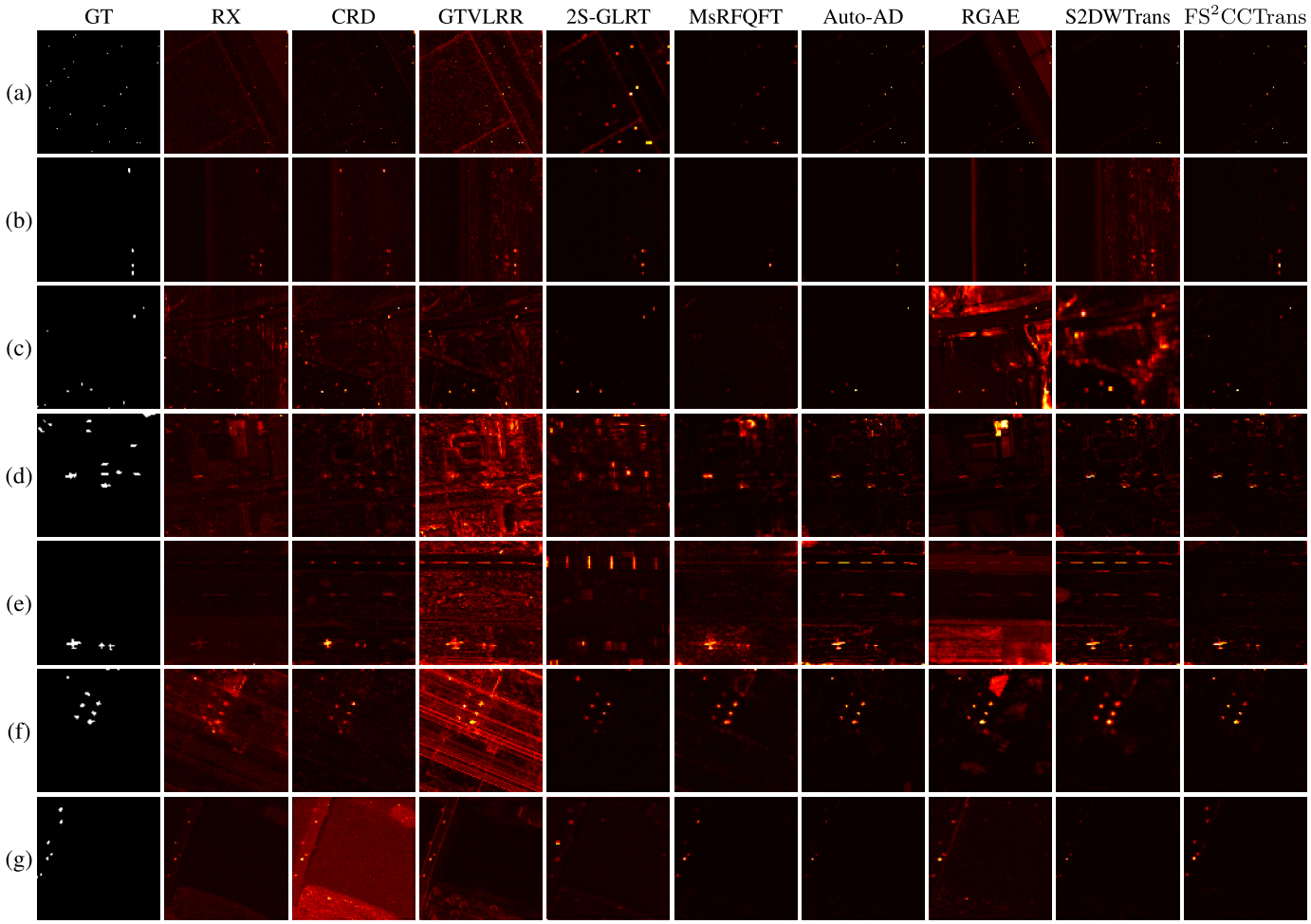


Fig. 12. Visualization of AD results. (a) Synthetic. (b) WHU-Hi-River. (c) HYDICE. (d) Los Angeles-1. (e) Gulfport. (f) Texas Coast. (g) Pavia.

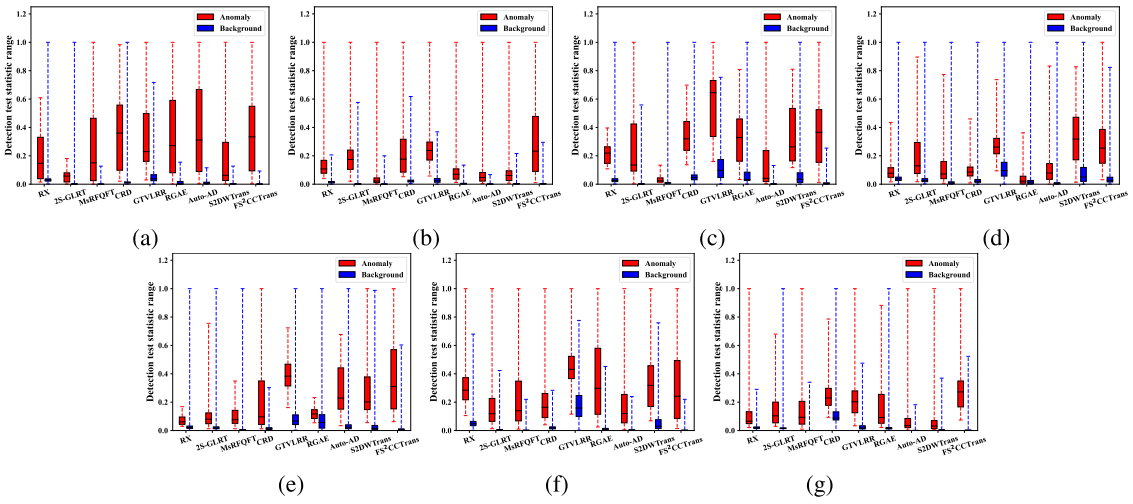


Fig. 13. Box-and-whisker plots of different methods on seven datasets. (a) Synthetic. (b) WHU-Hi-River. (c) HYDICE. (d) Los Angeles-1. (e) Gulfport. (f) Texas Coast. (g) Pavia.

robust detection in complex environments rather than marginal improvements on simpler cases. The consistent excellence of FS²CCTrans is further evidenced by AUC values exceeding 0.99 on the remaining six datasets, establishing its superior capability to maintain high detection sensitivity while effectively controlling false alarms.

3) Model Complexity: Table IV records the corresponding running time. RX, CRD, GTVLRR, 2S-GLRT, MsRFQFT, and RGAE are implemented on an Intel[4] Core[5]

[4]Registered trademark.
[5]Trademarked.

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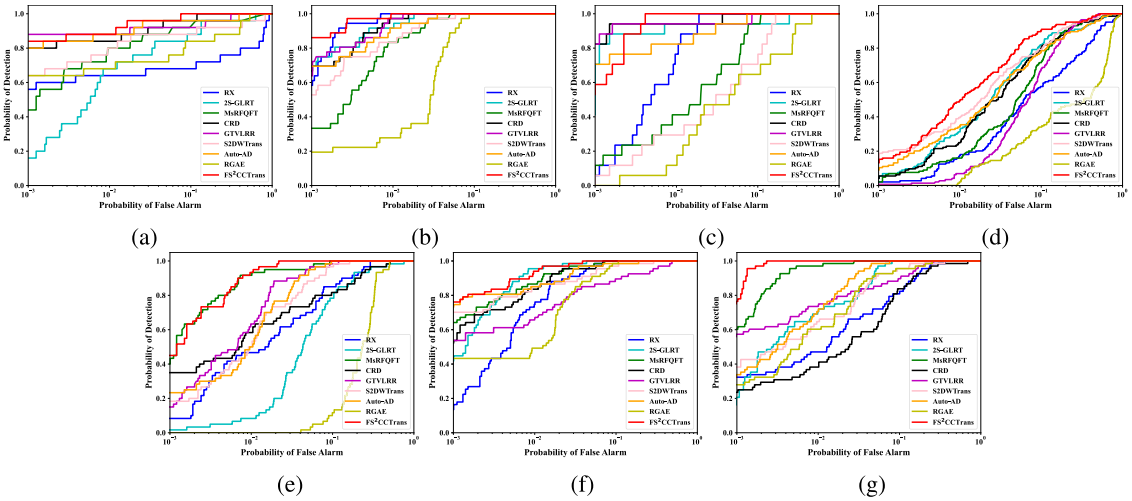


Fig. 14. ROC curves of different methods on the seven datasets. (a) Synthetic. (b) WHU-Hi-River. (c) HYDICE. (d) Los Angeles-1. (e) Gulfport. (f) Texas Coast. (g) Pavia.

TABLE III
AUC VALUES OF THE NINE DETECTORS ON SEVEN DATASETS

Dataset	RX	CRD	GTVLRR	2S-GLRT	MsRFQFT	Auto-AD	RGAE	S2DWTrans	FS^{2}CCTrans
Synthetic	0.8073	0.9913	0.9807	0.9491	0.9651	0.9822	0.9291	0.9431	**0.9957**
WHU-Hi-River	0.9988	0.9980	0.9982	0.9980	0.9935	0.9968	0.9722	0.9948	**0.9993**
HYDICE	0.9932	0.9975	0.9948	0.9846	0.9693	0.9924	0.8958	0.9450	**0.9988**
Los Angeles-1	0.8221	0.9212	0.9021	0.9236	0.9114	0.9046	0.6386	0.8809	**0.9553**
Gulfport	0.9526	0.9750	0.9871	0.9183	0.9945	0.9841	0.7610	0.9789	**0.9971**
Texas Coast	0.9907	0.9940	0.9607	0.9970	0.9954	0.9927	0.9822	0.9432	**0.9978**
Pavia	0.9538	0.9407	0.9759	0.9868	0.9984	0.9909	0.9782	0.9779	**0.9995**
Average	0.9312	0.9740	0.9714	0.9653	0.9754	0.9777	0.8796	0.9520	**0.9919**

TABLE IV
RUNNING TIMES (IN SECONDS) OF THE NINE DETECTORS ON SEVEN DATASETS

Data Set	Detectors and running times								
	RX	CRD	GTVLRR	2S-GLRT	MsRFQFT	RGAE	Auto-AD	S2DWTrans	FS^{2}CCTrans
Synthetic	0.19	7.96	95.06	30.86	0.28	105.02	121.17 / 12.40	1093.54 / 49.28	389.66 / 28.71
WHU-Hi-River	0.05	20.13	106.48	21.89	0.19	98.67	87.86 / 11.90	1219.18 / 51.92	377.04 / 27.58
HYDICE	0.09	4.24	71.49	11.17	0.16	23.03	218.54 / 21.08	1185.58 / 48.40	366.20 / 26.41
Los Angeles-1	0.10	15.94	77.89	29.54	0.24	68.20	83.52 / 12.98	1237.84 / 53.86	393.94 / 30.59
Gulfport	0.07	33.17	72.11	58.09	0.21	64.44	58.89 / 8.49	1149.04 / 47.34	353.81 / 26.07
Texas Coast	0.08	2.81	81.39	26.34	0.23	67.42	91.08 / 11.26	1172.50 / 48.13	331.42 / 24.55
Pavia	0.05	3.47	130.28	30.88	0.29	105.45	349.10 / 26.99	5309.66 / 169.81	784.51 / 41.24
Average	0.09	12.53	90.67	29.82	0.23	76.03	144.31 / 15.01	1766.76 / 66.96	428.08 / 29.31

Auto-AD, S2DWTrans and FS^{2}CCTrans are run on both CPU and GPU devices. We list the computational time as CPU/GPU.

i7-12700KF CPU following their source codes. Auto-AD, S2DWTrans, and the proposed FS^{2}CCTrans are executed on an NVIDIA GeForce 3060 GPU, with their CPU running times also provided. RX achieves the shortest running time, but the detection performance is unsatisfactory. MsRFQFT stands out for its efficiency, achieving the second shortest computation time across all datasets. Since GTVLRR requires many iterative optimizations, thus, requires more time compared to other conventional algorithms. Among GPU-accelerated methods, Auto-AD demonstrates superior speed (144.31 s average) through its lightweight spectral-focused architecture, which avoids computationally intensive spatial–spectral fusion. The proposed FS^{2}CCTrans incurs a 14.3-s computational overhead compared to Auto-AD, attributable to its comprehensive multidomain feature extraction framework. However, it achieves 37% faster execution than the structurally similar S2DWTrans (428.08 versus 1766.76 s average), demonstrating the efficiency gains from our CCA mechanism. This performance-efficiency balance positions FS^{2}CCTrans as a practical solution for complex AD scenarios requiring both high accuracy and operational feasibility.

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TABLE V
NETWORK PARAMETERS (MILLIONS)/FLOPS (BILLIONS) FOR DNN NETWORKS ON SEVEN DATASETS

Dataset	Auto-AD	RGAE	S2DWTrans	FS^{2}CCTrans
Synthetic	3.25 / 8.59	0.04 / 0.12	0.52 / 56.29	0.39 / 18.33
WHUHi-River	3.15 / 9.92	0.04 / 0.09	0.51 / 58.35	0.37 / 17.43
HYDICE	3.19 / 4.59	0.04 / 0.08	0.51 / 56.47	0.37 / 17.80
Los Angeles-1	3.25 / 5.98	0.04 / 0.08	0.53 / 56.78	0.38 / 18.09
Gulfport	3.23 / 5.91	0.04 / 0.07	0.52 / 57.13	0.37 / 18.33
Texas Coast	3.23 / 5.91	0.04 / 0.09	0.53 / 59.84	0.39 / 19.10
Pavia	3.10 / 12.21	0.04 / 0.08	0.51 / 57.19	0.38 / 18.66

TABLE VI
AUC SCORES OF THE ABLATION STUDY ON EVALUATING THE EFFECTIVENESS OF THE SM

Datasets	base1	base1 +IAM	base1 +EAM	base1 +SM	Time IR
WHU-Hi-River	0.9952	0.9984	0.9979	0.9993	3%
HYDICE	0.9926	0.9975	0.9981	0.9988	4%
Los Angeles-1	0.9280	0.9412	0.9448	0.9553	4%
Gulfport	0.9854	0.9960	0.9967	0.9971	3%
Texas Coast	0.9903	0.9966	0.9919	0.9978	3%
Pavia	0.9889	0.9933	0.9919	0.9995	3%

Table V provides a detailed comparison of the memory usage and computational complexity of the four DNN-based HAD models. The RGAE employs a single layer AE, the most basic structure in the compared models, and thus has the lowest complexity. Auto-AD has the largest number of parameters due to stacking multiple fully connected layers and residual connections. While both S2DWTrans and FS^{2}CCTrans employ transformer-based frameworks, our proposed method achieves remarkable efficiency improvements through the CCA mechanism. Specifically, FS^{2}CCTrans reduces parameter counts by 28% and FLOPs by 68% compared to S2DWTrans, while maintaining superior detection accuracy. Taking the Los Angeles-1 dataset as an example, the number of parameters in the proposed FS^{2}CCTrans is only about 11.69% of that in Auto-AD, reducing about 28% of the parameters and 68% of the FLOPs compared to that in S2DWTrans, highlighting the efficiency of FS^{2}CCTrans.

E. Ablation Study

1) Effectiveness of the Saliency Map: The SM is generated by analyzing the difference between the background and anomaly representations in both the amplitude and phase spectrum. To evaluate the performance of the proposed SM along with its two components, IAM and EAM, ablation experiments are conducted across six real-world datasets. First, we establish a base model by deactivating SM in FS^{2}CCTrans, then gradually integrate IAM, EAM, and SM into the base model. For the sake of clarity, we refer to these models as the base1 model, the base1 + IAM model, the base1 + EAM model, and the base1 + SM model. Consistent training epochs, batch sizes, and learning rates are used for fairness.

As visualized in Fig. 15(a2, b1), the base1 model suffers from anomaly overfitting, erroneously reconstructing anomalous pixels (yellow boxes) due to unconstrained optimization. Fig. 15(a3) illustrates that the base1 + IAM model effectively highlights anomaly regions, producing higher residual responses than base1. The base1 + EAM model in (a4) shows a similar behavior. We comparatively analyze these two strategies in terms of background suppression and anomaly emphasis. Taking the Los Angeles-1 dataset as an example, the base1 + IAM model successfully identifies most of the anomalies. However, the background suppression effect of IAM is unsatisfactory, causing portions of the background to be highlighted as well. EAM relies entirely on the phase information, making it more suitable when the anomaly and background have similar spectral but different spatial distributions. Nevertheless, its uniform treatment of the amplitude values diminishes the brightness of anomalies, thereby reducing the salience of detected targets and occasionally leading to an incomplete display. SM integrates the anomaly emphasis capability of IAM with the background suppression of EAM, producing a pure reconstructed background, as shown in Fig. 15(b2), with detection results closely approximating the ground truth.

TABLE VII
COMPARISON OF AUC AND EFFICIENCY OF FS^{2}CCTrans AND BASE2 MODEL ON SIX REAL DATASETS

Datasets	base2	FS^{2}CCTrans	Param DR	FLOPs DR
WHU-Hi-River	0.9917	0.9993	68%	55%
HYDICE	0.9902	0.9988	69%	58%
Los Angeles-1	0.9454	0.9553	68%	58%
Gulfport	0.9928	0.9971	67%	57%
Texas Coast	0.9944	0.9978	68%	58%
Pavia	0.9875	0.9995	69%	59%

Table VI records the AUC scores of the ablation experiments. Models integrating anomaly suppression mechanisms achieve considerable AUC improvements over the base1 model, with the proposed SM strategy attaining peak performance across all datasets. The base1 + IAM model demonstrates superior performance in the WHU-Hi-River, Texas Coast, and Pavia datasets, primarily attributable to the pronounced spectral differences between anomalous and background targets within these scenes. The EAM strategy is more adept for the HYDICE, Los Angeles-1, and Gulfport datasets, as the anomalies and backgrounds in these scenes have obvious differences in spatial structure. Furthermore, running time after using the SM strategy increased only by 3%, but the AUC score improved considerably. The experimental results fully confirm the effectiveness of the SM in steering the model toward more accurate AD.

2) Effectiveness of the Criss-Cross Attention: The RCCTrans consists of two successive CCTrans encoders, each containing a CCA block. To evaluate the contribution of the CCA blocks, we replace the CCA in RCCTrans with the standard MSA, denoted as the base2 model, and perform the experiments on six real datasets. Table VII presents the AUC scores and efficiency of FS^{2}CCTrans with or without CCA in six datasets, where Para DR represents the reduction rate in GPU memory usage of FS^{2}CCTrans after

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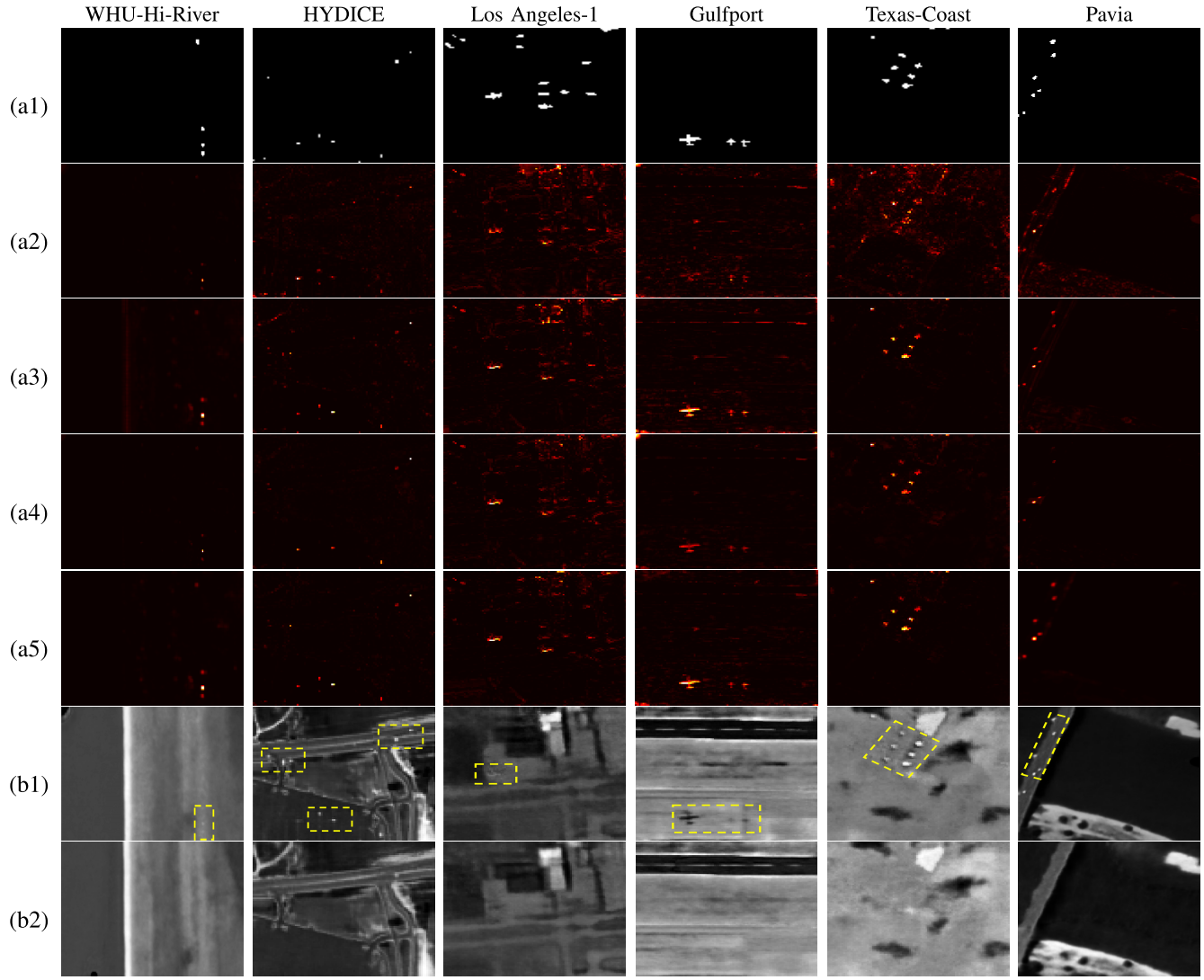


Fig. 15. Ablation visualization results on six datasets. (a1) Ground truth. (a2) base1 model. (a3) base1 + IAM model. (a4) base1 + EAM model. (a5) base1 + SM model. (b1) Reconstructed background of base1 model. (b2) Reconstructed background of base1 + SM model.

TABLE VIII

AUC SCORES (UPPER ROW) AND IMPROVEMENT ON NETWORKS BY THREE ANOMALY SUPPRESSION (LOWER ROW, COMPARED WITH THEIR ORIGINAL RESULTS) ON SIX REAL DATASETS

Dataset	AE				Auto-AD			FS^{2}CCTrans		
	Orig	$l_{2,1}$	AW	SM	$l_{2,1}$	AW	SM	$l_{2,1}$	AW	SM
WHU-Hi-River	0.9708 -	0.9715 +0.0007	0.9721 +0.0013	**0.9830** **+0.0122**	0.9972 +0.0004	0.9968 -	**0.9985** **+0.0017**	0.9984 -0.0009	0.9976 -0.0017	**0.9993** -
HYDICE	0.9701 -	0.9590 -0.0111	0.9668 -0.0033	**0.9803** **+0.0102**	0.9953 +0.0029	0.9924 -	**0.9979** **+0.0055**	0.9967 -0.0021	0.9960 -0.0028	**0.9988** -
Los Angeles-1	0.8181 -	0.8391 +0.0210	0.8282 +0.0101	**0.8416** **+0.0235**	0.8977 -0.0069	0.9046 -	**0.9348** **+0.0302**	0.9220 -0.0333	0.9354 -0.0199	**0.9553** -
Gulfport	0.9025 -	0.9252 +0.0227	0.9158 +0.0133	**0.9344** **+0.0319**	0.9666 -0.0175	0.9841 -	**0.9935** **+0.0094**	0.9806 -0.0165	0.9887 -0.0084	**0.9971** -
Texas Coast	0.9810 -	0.9836 +0.0026	0.9819 +0.0009	**0.9880** **+0.007**	0.9731 -0.0196	0.9927 -	**0.9963** **+0.0036**	0.9811 -0.0167	0.9955 -0.0023	**0.9978** -
Pavia	0.9401 -	0.9123 -0.0278	0.9317 -0.0084	0.9438 **+0.0037**	0.9911 +0.0002	0.9909 -	0.9917 **+0.0008**	0.9925 -0.0070	0.9917 -0.0078	0.9995 -
Average	-	+0.0014	+0.0023	**+0.0148**	-0.0067	-	**+0.0085**	-0.0127	-0.0071	-

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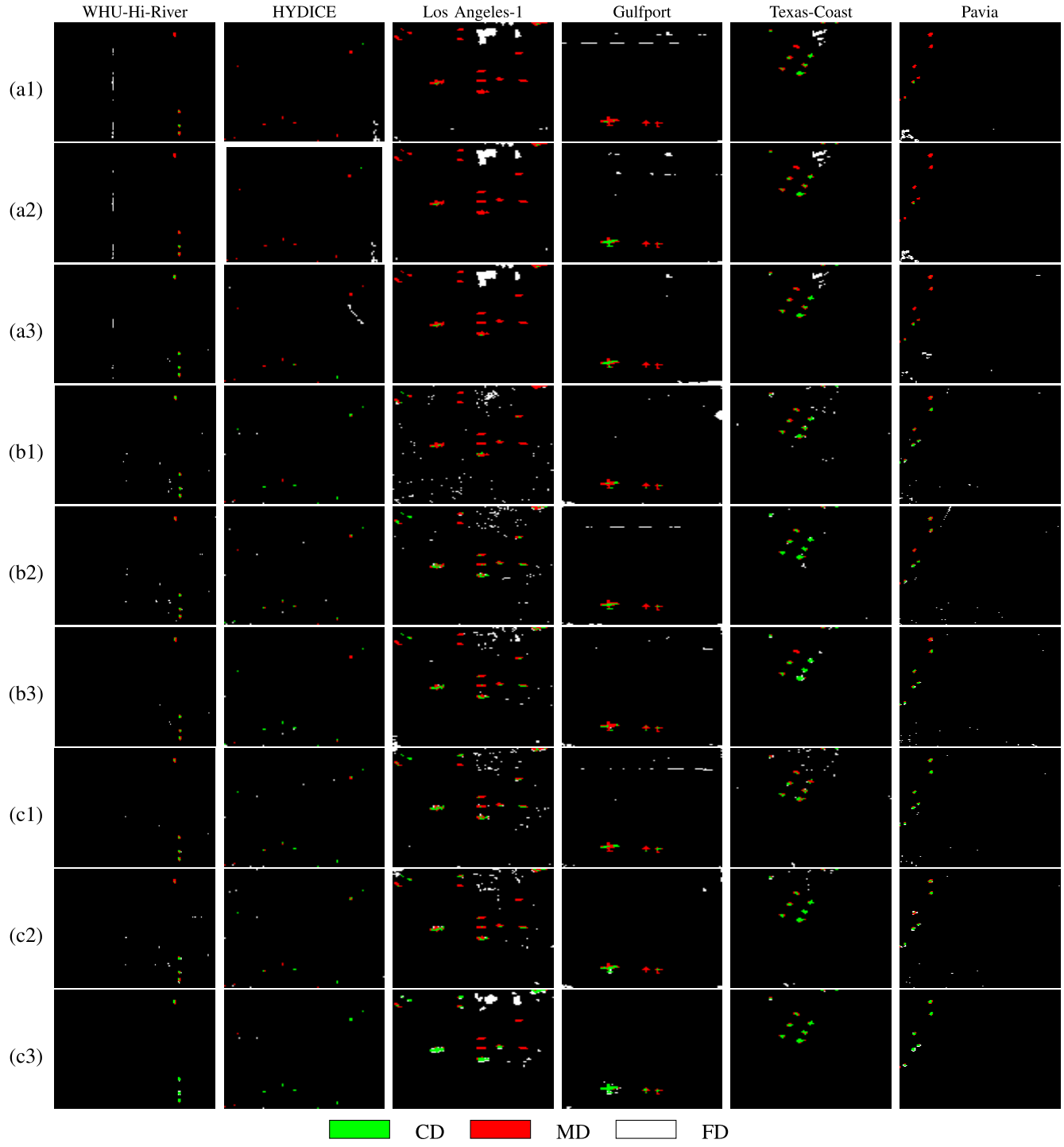


Fig. 16. Binary AD maps of three anomaly suppression strategies, $l_{2,1}$, AW, and SM, combined with AE, Auto-AD,and FS^{2}CCTrans networks. (a1) AE + $l_{2,1}$. (a2) AE + AW. (a3) AE + SM. (b1) Auto-AD + $l_{2,1}$. (b2) Auto-AD (AW). (b3) Auto-AD + SM. (c1) FS^{2}CCTrans + $l_{2,1}$. (c2) FS^{2}CCTrans + AW. (c3) FS^{2}CCTrans (SM).

using CCA, and FLOPs DR is the decrease rate of FLOPs. FS^{2}CCTrans consistently demonstrates superior precision in all datasets compared to the base2 mode. This enhancement is evident not only in the improved AUC scores but also in significant reductions in resource consumption, specifically approximately 68% in GPU memory and 58% in FLOPs. This efficiency stems from the CCA's strategic processing of spatial and spectral information, which requires less computational overhead by capturing local dependencies in a criss-cross manner and subsequently aggregates global information through recursive operations. This multistep aggregation results in a more robust feature representation, which helps to reduce false positives and consequently higher AUC values. The optimized computational demand and reduced memory footprint make FS^{2}CCTrans suitable for complex AD tasks in resource-constrained environments.

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3) Transferability Discussion: To verify the generalizability and effectiveness of the SM, we assess the performance improvement of a network employing this method versus the baseline setup. Additionally, we evaluate two other anomaly suppression methods: adaptive weighted training (AW) in Auto-AD and the $l_{2,1}$-norm in RGAE. In subsequent tests, these three anomaly suppression methods are used in the proposed FS^{2}CCTrans, Auto-AD, and AE[6] models, leading to nine trials that assess the SM's transferability and its performance compared to other suppression strategies (For example, in the Auto-AD model experiments, we substitute its AW method with $l_{2,1}$ and SM individually, instead of combining the strategies). The training protocols for Auto-AD and AE followed those specified in the original publications.

Table VIII shows the AUC scores for all combinations on six real datasets. To facilitate a precise comparison, we also report the enhancement scores of three anomaly suppression strategies over the baseline methods: the AE detector using the squared Frobenius norm of the reconstruction error as the objective, Auto-AD, and the newly introduced FS^{2}CCTrans. The dataset-specific highest AUC scores and most pronounced improvements are highlighted in bold, while original AUC scores are shown in gray. Notably, our SM strategy consistently shows the greatest improvements in all cases. For instance, on the Los Angeles-1 dataset, integrating SM with AE networks boosts the AUC from 0.8181 to 0.8416. Relative to the AW strategy, utilizing SM enhances the Auto-AD AUC score by 3%. Generally, AW's performance slightly surpasses that of $l_{2,1}$. The AW anomaly suppression principle resembles that of SM, where scaling down the reconstruction error allows the network to decrease weights for anomaly pixels in the loss function, thus reducing their reconstruction during training. In contrast, the $l_{2,1}$ norm focuses on local gradient adjustments for individual samples, which might not be adequately detailed for addressing globally dispersed anomalies.

Binary detection maps are utilized for qualitatively evaluating detector performance. Fig. 16 illustrates the generated binary detection maps for all combinations by segmenting the grayscale detection maps based on specified thresholds. These maps simultaneously show correct detection (CD), MD, and false detection (FD), where CD accurately identifies anomaly pixels, MD misclassifies anomaly pixels as background, and FD errors background pixels as anomalies. Here, the threshold is calculated by the anomaly pixel ratio in the image. Table I shows the WHU-Hi-River dataset has a 0.2% anomaly ratio, yielding a threshold of 99.8%, with other datasets having thresholds of 99.8%, 98.56%, 99.4%, 99.33%, and 98.7% for HYDICE, Los Angeles-1, Gulfport, Texas Coast, and Pavia, respectively. Fig. 16 evidences that integrating the proposed suppression mechanism into AE, Auto-AD, and FS^{2}CCTrans detectors considerably reduces missed and FDs more effectively than $l_{2,1}$ and AW strategies. Importantly, the AD results of FS^{2}CCTrans closely match the ground truth, demonstrating that the novel suppression strategy excels in transferability and optimized anomaly target suppression.

[6]To eliminate the confounding effect of parameters, the graph regularization component was removed from the RGAE model, retaining only its AE network.

V. Conclusion

This study focuses on the discriminability between background and anomalous targets in the HAD task. We propose FS^{2}CCTrans, an efficient HAD algorithm, which provides new insights for fully and rationally exploiting the unique characteristics of background and anomalous targets in the frequency–spatial–spectral domain in the AD task. Specifically, in the frequency domain, we analyze the representational properties of the background and anomalous targets in the amplitude and phase spectrum to obtain the SM in a nonparametric learning manner. This SM assists in training the network without increasing computational complexity, guiding it in learning the background and suppressing anomalies. In the spatial–spectral domain, we designed a feature extraction module based on criss-cross attention, called CCTrans, to deeply mine the correlation between background and abnormal targets, thereby better separating them. The criss-cross attention module can capture the whole image context information with less computational and memory costs, significantly reducing the computational complexity of the FS^{2}CCTrans. Comparison experiments with eight state-of-the-art methods on synthetic and real datasets demonstrate the superiority of the FS^{2}CCTrans.

References

[1] B. Zhang et al., "Progress and challenges in intelligent remote sensing satellite systems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1814–1822, 2022.

[2] C.-I. Chang, "Hyperspectral target detection: Hypothesis testing, signal-to-noise ratio, and spectral angle theories," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505223.

[3] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[4] C.-I. Chang, "Target-to-anomaly conversion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540428.

[5] L. Gao, X. Sun, X. Sun, L. Zhuang, Q. Du, and B. Zhang, "Hyperspectral anomaly detection based on chessboard topology," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505016.

[6] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[7] M. Xu et al., "Regionally and locally adaptive models for retrieving chlorophyll—A concentration in inland waters from remotely sensed multispectral and hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4758–4774, Jul. 2019.

[8] B. Lu, P. Dao, J. Liu, Y. He, and J. Shang, "Recent advances of hyperspectral imaging technology and applications in agriculture," *Remote Sens.*, vol. 12, no. 16, p. 2659, Aug. 2020.

[9] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Aug. 1990.

[10] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[11] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

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[12] J. Liu, Z. Hou, W. Li, R. Tao, D. Orlando, and H. Li, "Multipixel anomaly detection with unknown patterns for hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5557–5567, Oct. 2022.

[13] B. Tu, X. Yang, W. He, J. Li, and A. Plaza, "Hyperspectral anomaly detection using reconstruction fusion of quaternion frequency domain analysis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 8358–8372, Jun. 2023.

[14] L. Gao, Z. Wang, L. Zhuang, H. Yu, B. Zhang, and J. Chanussot, "Using low-rank representation of abundance maps and nonnegative tensor factorization for hyperspectral nonlinear unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504017.

[15] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

[16] L. Zhuang, M. K. Ng, and Y. Liu, "Cross-track illumination correction for hyperspectral pushbroom sensor images using low-rank and sparse representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502117.

[17] Q. Ling, Y. Guo, Z. Lin, and W. An, "A constrained sparse representation model for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2358–2371, Apr. 2019.

[18] Z. Hou, W. Li, R. Tao, P. Ma, and W. Shi, "Collaborative representation with background purification and saliency weight for hyperspectral anomaly detection," *Sci. China Inf. Sci.*, vol. 65, no. 1, pp. 1–12, Jan. 2022.

[19] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[20] R. Feng, H. Li, L. Wang, Y. Zhong, L. Zhang, and T. Zeng, "Local spatial constraint and total variation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512216.

[21] S. Lin, M. Zhang, X. Cheng, K. Zhou, S. Zhao, and H. Wang, "Dual collaborative constraints regularized low-rank and sparse representation via robust dictionaries construction for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2009–2024, 2023.

[22] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[23] G. Zhang, N. Li, B. Tu, Z. Liao, and Y. Peng, "Hyperspectral anomaly detection via dual collaborative representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4881–4894, 2020.

[24] S. Feng, D. Wu, R. Feng, and C. Zhao, "Hyperspectral anomaly detection with total variation regularized low rank tensor decomposition and collaborative representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[25] Z. Wang, M. K. Ng, L. Zhuang, L. Gao, and B. Zhang, "Nonlocal self-similarity-based hyperspectral remote sensing image denoising with 3-D convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531617.

[26] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, and Q. Li, "Hyperspectral anomaly detection via deep plug-and-play denoising CNN regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9553–9568, Nov. 2021.

[27] X. Lu, W. Zhang, and J. Huang, "Exploiting embedding manifold of autoencoders for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1527–1537, Nov. 2020.

[28] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.

[29] W. Rao, Y. Qu, L. Gao, X. Sun, Y. Wu, and B. Zhang, "Transferable network with Siamese architecture for anomaly detection in hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102669.

[30] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2020, pp. 1–12.

[31] J. Li, M. D. Levine, X. An, X. Xu, and H. He, "Visual saliency based on scale-space analysis in the frequency domain," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 4, pp. 996–1010, Apr. 2013.

[32] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

[33] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314.

[34] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314.

[35] P. Xiang, S. Ali, S. K. Jung, and H. Zhou, "Hyperspectral anomaly detection with guided autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538818.

[36] T. Jiang, W. Xie, Y. Li, J. Lei, and Q. Du, "Weakly supervised discriminative learning with spectral constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6504–6517, Nov. 2022.

[37] D. Wang, L. Gao, Y. Qu, X. Sun, and W. Liao, "Frequency-to-spectrum mapping GAN for semisupervised hyperspectral anomaly detection," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1258–1273, Dec. 2023.

[38] S. Xiao, T. Zhang, Z. Xu, J. Qu, S. Hou, and W. Dong, "Anomaly detection of hyperspectral images based on transformer with spatial–spectral dual-window mask," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1414–1426, 2023.

[39] B. Kumar, O. Dikshit, A. Gupta, and M. K. Singh, "Feature extraction for hyperspectral image classification: A review," *Int. J. Remote Sens.*, vol. 41, no. 16, pp. 6248–6287, 2020.

[40] X. Zhao, W. Li, T. Shan, L. Li, and R. Tao, "Hyperspectral target detection by fractional Fourier transform," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 1655–1658.

[41] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, Dec. 2014, pp. 58–62.

[42] M. S. Stefanou and J. P. Kerekes, "A method for assessing spectral image utility," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 6, pp. 1698–1706, Jun. 2009.

[43] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.



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